Ex.No.4

Program :

```
#include <stdio.h>

int main()
{
   int bt[20], p[20], wt[20], tat[20];
   int i, j, n, total = 0, pos, temp;
   float avg_wt, avg_tat;

   printf("Enter number of process:");
   scanf("%d", &n);

   printf("\nEnter Burst Time:\n");
   for (i = 0; i < n; i++)
   {
      printf("p%d:", i + 1);
      scanf("%d", &bt[i]);
      p[i] = i + 1;
   }

   for (i = 0; i < n; i++)
   {
      pos = i;
      for (j = i + 1; j < n; j++)
      {
         if (bt[j] < bt[pos])
            pos = j;
      }

      temp = bt[i];
```

```
    bt[i] = bt[pos];
    bt[pos] = temp;

    temp = p[i];
    p[i] = p[pos];
    p[pos] = temp;
}

wt[0] = 0;

for (i = 1; i < n; i++)
{
    wt[i] = 0;
    for (j = 0; j < i; j++)
        wt[i] += bt[j];

    total += wt[i];
}

avg_wt = (float)total / n;
total = 0;

printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");

for (i = 0; i < n; i++)
{
    tat[i] = bt[i] + wt[i];
    total += tat[i];

    printf("\np%d\t\t%d\t\t%d\t\t%d", p[i], bt[i], wt[i], tat[i]);
}
```

```
    avg_tat = (float)total / n;

    printf("\n\nAverage Waiting Time=%f", avg_wt);
    printf("\nAverage Turnaround Time=%f\n", avg_tat);

    return 0;
}
```

Output Clear

```
Enter number of processes: Enter number of processes: 4



Enter Burst Time and Priority for Process P1: 5 2

Enter Burst Time and Priority for Process P2: 3 1

Enter Burst Time and Priority for Process P3: 8 4

Enter Burst Time and Priority for Process P4: 6 3

Process Burst Time  Priority    Waiting Time    Turnaround Time


=== Code Execution Successful ===
```